

GRAND MARAIS AP, MN, USA

WMO: 727454

Lat: 47.838N

Lon: 90.383W

Elev: 1798

StdP: 13.77

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04911

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.3	-14.7	-27.3	1.3	-16.4	-23.9	1.6	-10.9	25.3	11.7	22.8	8.6	9.2	320	0.629

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	80.6	66.3	76.9	63.4	73.4	61.7	68.9	76.8	66.7	73.7	64.4	70.4	7.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	102.3	73.8	63.8	94.6	70.4	62.6	90.8	68.8	34.1	77.4	32.2	73.3	30.6	71.0	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.5	18.7	16.6	DB	-24.3	86.1	4.3	2.8	-27.4	88.1	-29.9	89.8	-32.3	91.4	-35.4	93.5	(4)
(5)			WB	-24.0	72.4	4.5	2.8	-27.2	74.4	-29.9	76.1	-32.4	77.7	-35.7	79.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.0	9.0	12.0	23.3	35.4	47.7	56.7	62.3	60.6	53.0	41.0	27.5	14.5		
	DBStd	20.99	12.89	12.38	12.33	9.23	8.32	6.55	6.05	5.98	7.92	8.52	10.04	12.40		
	HDD50	5908	1270	1064	828	444	145	15	1	2	58	302	677	1102		
	HDD65	10290	1735	1484	1291	888	540	259	127	163	366	744	1126	1567		
	CDD50	1179	0	0	2	6	73	215	381	330	148	23	1	0		
	CDD65	85	0	0	0	0	3	9	42	26	5	0	0	0		
	CDH74	617	0	0	0	1	29	74	305	170	35	3	0	0		
	CDH80	72	0	0	0	0	2	6	39	22	3	0	0	0		
	WSAvg	6.9	8.0	7.9	7.4	7.6	7.2	6.0	5.5	5.4	5.8	7.0	7.4	7.9		
	PrecAvg	26.7	1.1	0.8	1.1	2.0	2.7	3.6	3.3	2.8	3.1	3.3	2.0	1.4		
	PrecMax	35.7	2.4	1.9	3.3	7.4	4.8	8.3	8.0	5.8	6.8	8.7	3.9	3.0		
(15) Precipitation	PrecMin	20.4	0.3	0.2	0.4	0.3	0.8	0.4	1.3	0.9	0.8	1.3	0.6	0.3		
	PrecStd	4.3	0.6	0.5	0.7	1.5	1.0	1.8	1.7	1.2	1.4	1.9	0.9	0.7		
	DB	37.2	43.1	57.2	66.3	78.7	80.7	83.7	82.3	79.1	70.1	54.4	40.2			
	MCWB	32.9	36.6	48.5	50.7	60.1	66.7	67.7	68.2	62.8	58.3	47.2	36.1			
	2% DB	33.8	37.0	49.6	60.5	72.5	75.3	81.0	78.9	72.9	63.4	48.3	36.8			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	30.9	33.9	41.9	46.8	56.8	61.2	66.9	65.7	62.8	53.6	43.8	33.7			
	5% DB	29.9	34.2	44.8	54.6	67.7	72.7	77.4	74.9	70.0	57.5	44.9	34.1			
	MCWB	28.6	31.2	38.4	43.2	54.3	59.9	64.4	63.5	61.5	50.8	40.1	31.8			
	10% DB	27.2	29.8	38.9	50.5	63.1	70.0	74.3	72.3	65.8	54.4	41.1	30.5			
	MCWB	25.3	27.4	33.9	40.8	51.7	58.9	62.9	61.9	58.9	48.7	37.1	29.2			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	33.9	37.4	51.3	54.4	64.3	69.2	72.8	71.8	68.2	60.5	50.2	37.0			
	MCDB	36.9	40.2	55.3	61.9	70.7	77.4	80.3	79.3	74.5	65.3	52.9	39.2			
	2% WB	30.5	34.9	43.5	48.2	59.7	65.7	69.5	68.0	64.8	56.2	44.5	34.4			
	MCDB	32.5	37.4	48.3	58.0	67.2	72.3	77.4	76.1	69.6	61.5	47.6	36.7			
	5% WB	28.3	31.0	38.2	44.3	56.4	62.8	67.0	65.9	62.4	52.2	40.4	31.8			
(31) Mean Daily Temperature Range	MCDB	29.4	33.9	43.3	53.0	64.6	68.3	74.3	72.6	67.2	57.4	44.2	33.9			
	10% WB	25.1	27.4	34.5	41.1	53.0	60.2	64.7	63.7	59.6	49.2	37.4	29.8			
	MCDB	27.1	29.0	39.2	49.4	61.2	67.0	71.5	69.3	65.1	54.4	40.6	31.1			
	5% DB	15.0	15.6	19.1	25.2	26.8	23.2	21.7	21.8	22.1	20.2	15.7	13.4			
	MCWBR	13.2	12.6	13.2	15.3	14.3	13.1	12.0	12.4	13.7	13.4	11.9	11.8			
(38) Clear-Sky Solar Irradiance	5% WB	13.9	14.1	17.8	23.0	21.9	18.9	19.5	19.4	18.1	18.1	14.8	12.8			
	MCWBR	12.6	11.7	13.0	15.1	13.3	12.9	12.4	12.6	12.7	13.4	11.9	11.6			
	taub	0.245	0.254	0.286	0.325	0.342	0.363	0.372	0.358	0.342	0.301	0.276	0.251			
	taud	2.434	2.410	2.428	2.425	2.458	2.416	2.433	2.490	2.508	2.594	2.558	2.466			
	Ebn at Noon	273	294	299	295	292	285	281	281	276	274	260	257			
(44) All-Sky Solar Radiation	Edh at Noon	20	26	30	34	34	36	35	31	28	22	18	17			
	RadAvg	383	706	1114	1492	1709	1833	1908	1622	1186	708	383	292			
	RadStd	49	78	102	154	114	134	88	114	79	84	47	28			

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page